

Paper 2 - Set 02 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	B	Differentiation
2	C	Polynomials, factor and remainder theorems
3	A	Graphs of functions
4	H	Sequences and series
5	A	Triangles, radians and sectors
6	A	Sequences and series
7	D	Chains of reasoning
8	B	Chains of reasoning
9	F	Chains of reasoning
10	C	The logic of arguments
11	C	Mathematical proof
12	F	Differentiation
13	A	Mathematical proof
14	E	The logic of arguments
15	E	Identifying errors in proofs
16	F	Quadratic functions
17	E	Triangles, radians and sectors
18	B	Graphs of functions
19	B	The logic of arguments
20	E	Mathematical proof

Worked solutions

Question 1 DIF-17 · Differentiation · spec MM6.3

The curve $y = x^2 + kx$ (with $k < 0$) crosses the x -axis at the origin and at one other point P . The gradient of the curve at P is 6. Find k .

Answer: B

$$P = (-k, 0). y' = 2x + k = -2k + k = -k = 6, \text{ so } k = -6.$$

Full method - how and why

Find the second crossing point in terms of k , then impose the gradient condition there.

- 1 Find the x -intercepts: $x^2 + kx = x(x + k) = 0$, so $x = 0$ or $x = -k$. Thus $P = (-k, 0)$.

why: With $k < 0$, $-k$ is positive - the curve crosses to the right of the origin.

- 2 Differentiate: $y' = 2x + k$.

why: The gradient at any point.

- 3 Evaluate at P : $y'(-k) = -2k + k = -k$.

why: Substituting $x = -k$.

- 4 Set equal to 6: $-k = 6$, so $k = -6$.

why: Consistent with the given condition $k < 0$ ✓.

Where it goes wrong

- Using $x = k$ for the second intercept, giving $k = 2$ or $3k = 6$.
- Answering 6 and ignoring the stated sign of k .

Takeaway Find the point in terms of the parameter first, then apply the gradient condition; check any sign restriction at the end.

Question 2 POL-07 · Polynomials, factor and remainder theorems · spec MM1.6

Find the quotient when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 1)$.

Answer: C

$$2x^3 - 3x^2 + 4x - 5 = (x - 1)(2x^2 - x + 3) - 2; \text{ check the expansion}$$

$$2x^3 - x^2 + 3x - 2x^2 + x - 3 = 2x^3 - 3x^2 + 4x - 3, \text{ then subtract 2 for the remainder.}$$

Full method - how and why

A division question where only the quotient is wanted. Comparing coefficients is quicker and safer than long division.

- 1 Write $2x^3 - 3x^2 + 4x - 5 = (x - 1)(Ax^2 + Bx + C) + R$.

why: Dividing a cubic by a linear factor gives a quadratic quotient and a constant remainder.

- 2 Compare x^3 terms: $A = 2$.

why: Only one product contributes to the highest power, so the leading coefficient comes free.

- 3 Compare x^2 terms: $B - A = -3$, so $B = -1$.

why: The x^2 term comes from $x \cdot Bx$ and $-1 \cdot Ax^2$.

- 4 Compare x terms: $C - B = 4$, so $C = 3$. The quotient is $2x^2 - x + 3$.

why: The x term comes from $x \cdot C$ and $-1 \cdot Bx$. (The constant term then gives $R = -2$, but the question does not need it.)

Where it goes wrong

- Getting the sign of B wrong, giving $2x^2 + x + 3$ - which is offered.
- Reporting the remainder instead of the quotient.

Takeaway Comparing coefficients from the top down finds a quotient one coefficient at a time, with no long-division layout to get wrong.

Question 3 GRA-10 · Graphs of functions · spec MM8.6

A polynomial of degree 5 with real coefficients has how many real roots at least, and at most?

Answer: A

An odd-degree polynomial tends to opposite infinities at each end, so it crosses the axis at least once; a degree-5 polynomial has at most 5 roots.

Full method - how and why

Two standard facts about polynomial roots: the maximum is the degree, and odd degree forces at least one real root.

- 1 At most: a degree-5 polynomial has at most 5 roots, since each root contributes a linear factor.
why: More than 5 factors would raise the degree above 5.
- 2 At least: as $x \rightarrow +\infty$ and $x \rightarrow -\infty$, an odd-degree polynomial tends to opposite infinities.
why: The leading term ax^5 dominates and changes sign with x .
- 3 A continuous function that is positive somewhere and negative somewhere must cross zero, so there is at least one real root.
why: Intermediate value reasoning; this is why every odd-degree polynomial has a real root.
- 4 At least 1, at most 5.
why: Match to the options.

Where it goes wrong

- Saying 'at least 0' by analogy with even degrees like $x^2 + 1$, which genuinely have none.
- Saying 'exactly 5', which counts complex roots or ignores repeats.

Takeaway Odd degree $\Rightarrow$ at least one real root; degree $n \Rightarrow$ at most n real roots.

Question 4 X2-23 · Sequences and series · spec MM2.2

An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?

- I If $d > 0$ then every term after some point is positive.
- II The sum of the first n terms is a quadratic function of n .
- III If a and d are integers then S_n is an integer for every n .

Answer: H

I: terms $a + (n - 1)d$ eventually exceed any bound. II: $S_n = \frac{n}{2}(2a + (n - 1)d)$ is quadratic in n . III: S_n is a sum of integers.

Full method - how and why

Three claims about an arithmetic sequence, settled by the n th-term and sum formulas.

- 1 I: the terms are $a + (n - 1)d$ with $d > 0$, which increases without bound, so beyond some point every term is positive. **True.**
why: A positive common difference eventually overcomes any negative start.
- 2 II: $S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{d}{2}n^2 + (a - \frac{d}{2})n$ - a quadratic in n (with $d \neq 0$, so the n^2 term survives). **True.**
why: Expand and collect powers of n ; the hypothesis $d \neq 0$ is what makes it genuinely quadratic.
- 3 III: S_n is a sum of n integers $a, a + d, \dots$, hence an integer. **True.**
why: Sums of integers are integers - no need to worry about the $\frac{n}{2}$ in the formula, since the bracket is even when n is odd and vice versa.
- 4 So all three.
why: Match to the list.

Where it goes wrong

- Rejecting III because the formula contains $\frac{n}{2}$ - the product still comes out whole.
- Rejecting II if d were 0 (linear) - but the question states $d \neq 0$.
- Reading I as 'every term is positive' rather than 'every term after some point'.

Takeaway S_n for an arithmetic series is quadratic in n ; check the stated conditions (a, d non-zero) before dismissing a claim.

Question 5 X2-15 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , P : ' $a^2 = b^2 + c^2$ '. Q : 'angle $A = 90^\circ$ '. Which one of the following is true?

Answer: A

Pythagoras and its converse: the cosine rule gives $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, which is 0 exactly when $a^2 = b^2 + c^2$.

Full method - how and why

Pythagoras **and** its converse both hold, so this is a genuine equivalence - best seen through the cosine rule.

1 The cosine rule gives $a^2 = b^2 + c^2 - 2bc \cos A$, so $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$.

why: Rearranging the cosine rule turns both statements into one equation.

2 $a^2 = b^2 + c^2$ makes the numerator zero, so $\cos A = 0$ and (in a triangle) $A = 90^\circ$.

why: $2bc \neq 0$ since side lengths are positive; and $0^\circ < A < 180^\circ$ makes $A = 90^\circ$ the only solution of $\cos A = 0$.

3 Conversely $A = 90^\circ$ gives $\cos A = 0$, so $b^2 + c^2 - a^2 = 0$.

why: The same equation, read the other way.

4 So P and Q are equivalent: necessary and sufficient.

why: This is Pythagoras together with its converse.

Where it goes wrong

- Assuming only the forward direction (right angle $\Rightarrow$ Pythagoras) and answering 'necessary but not sufficient'.
- Forgetting that a must be the side **opposite** A for the relation to hold.

Takeaway The cosine rule contains Pythagoras and its converse: $\cos A = 0 \Leftrightarrow a^2 = b^2 + c^2$, with a opposite A .

Question 6 X2-04 · Sequences and series · spec MM2.3

A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?

Answer: A

A geometric series converges if and only if $|r| < 1$, which is exactly $-1 < r < 1$.

Full method - how and why

A definition-driven case: the convergence condition for a geometric series is exactly the stated inequality, so both directions hold.

1 A geometric series with $a \neq 0$ has a sum to infinity if and only if $|r| < 1$.

why: The partial sums $\frac{a(1-r^n)}{1-r}$ converge precisely when $r^n \rightarrow 0$.

2 $|r| < 1$ is the same statement as $-1 < r < 1$.

why: Modulus inequalities unfold into a double inequality.

3 So P and Q are equivalent: necessary **and** sufficient.

why: An 'if and only if' condition gives both directions at once.

Where it goes wrong

- Forgetting the case $r = \pm 1$: $r = 1$ gives partial sums na , which diverge; $r = -1$ oscillates. Both are correctly excluded.
- Answering 'sufficient but not necessary' from habit - here no counterexample exists.
- Overlooking why $a \neq 0$ is stated: with $a = 0$ every r would give a (trivial) sum.

Takeaway Geometric series converge $\Leftrightarrow |r| < 1$, and then $S_\infty = \frac{a}{1-r}$. An 'if and only if' condition is necessary and sufficient.

Question 7 RSN-22 · Chains of reasoning · spec Prf5

A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?

Answer: D

Each of the 4 rows contains exactly 2 black squares: 8 in total.

Full method - how and why

Count by rows - the column condition is extra information, not needed for the count.

- 1 There are 4 rows and each contains exactly 2 black squares.

why: Every black square lies in exactly one row, so rows partition them.

- 2 Total $= 4 \times 2 = 8$.

why: No double counting.

- 3 Check with the columns: $4 \times 2 = 8$ as well - consistent, which is why such a colouring can exist.

why: If the two counts disagreed, no such grid would exist.

Where it goes wrong

- Adding rows and columns to get 16 - that counts every black square twice.
- Answering 4 by counting rows rather than squares.

Takeaway Count via one partition only; use the other as a consistency check, not as extra squares.

Question 8 RSN-07 · Chains of reasoning · spec Prf5

The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?

Answer: B

The three rows together contain every digit once, so $3 \times (\text{row sum}) = 1 + 2 + \dots + 9 = 45$, giving 15.

Full method - how and why

Count the same total two different ways - the standard double-counting move.

- 1 The three rows between them contain each digit 1 to 9 exactly once.

why: Every cell lies in exactly one row, so the rows partition the grid.

- 2 So the three row sums add to $1 + 2 + \dots + 9 = 45$.

why: Use $\frac{9 \times 10}{2} = 45$.

- 3 All three row sums are equal, so each is $\frac{45}{3} = 15$.

why: And the columns give the same total, so they must each be 15 too - consistent.

Where it goes wrong

- Answering 45, the grand total rather than one row's sum.
- Trying to construct an actual magic square; the total settles it with no construction needed.

Takeaway Double-count a total: rows partition the grid, so $(\text{number of rows}) \times (\text{row sum}) = \text{grand total}$.

Question 9 RSN-16 · Chains of reasoning · spec Prf5

The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?

Answer: F

$(m - 4) + (m - 2) + m + (m + 2) + (m + 4) = 5m$ where m is the middle (odd) number; $5m$ is odd, so 10 is never a factor.

Full method - how and why

Represent the numbers symmetrically about the middle so everything cancels, then read off the guaranteed factor.

- 1 Let the middle number be m (odd). The five are $m - 4, m - 2, m, m + 2, m + 4$.
why: Consecutive odd numbers differ by 2; centring makes the offsets cancel in pairs.
- 2 Add: the ± 2 and ± 4 cancel, leaving $S = 5m$.
why: So 5 is always a factor.
- 3 Is 10 always a factor? m is odd, so $5m$ is odd and never divisible by 2. **No.**
why: For instance $1 + 3 + 5 + 7 + 9 = 25$.
- 4 Are 3, 15 or 25 always factors? 25 needs $5 \mid m$, 3 needs $3 \mid m$ - neither is guaranteed ($m = 7$ gives $S = 35$).
why: Only the factor 5 survives every case.

Where it goes wrong

- Answering 25 from the single example $1 + 3 + 5 + 7 + 9 = 25$ - that is $5m$ with $m = 5$, a coincidence.
- Answering 10 by assuming an even total; five odd numbers sum to an odd number.

Takeaway An odd count of equally spaced terms sums to (count) $\times$ (middle term). Test the extra factors against a second example.

Question 10 LGC-04 · The logic of arguments · spec Arg1

Consider the statement about a real number x : **if** $x^2 = x$, **then** $x = 1$. Which one of the following is true?

Answer: C

$x = 0$ satisfies $x^2 = x$ but not $x = 1$, so the statement is false. The converse, 'if $x = 1$ then $x^2 = x$ ', is true.

Full method - how and why

Judge a statement and its converse separately. Each needs either a proof or a counterexample.

- 1 The statement: if $x^2 = x$ then $x = 1$. Solve $x^2 = x$: $x(x - 1) = 0$, so $x = 0$ or $x = 1$.
why: Never divide by x - that is exactly how the root $x = 0$ gets lost.
- 2 $x = 0$ satisfies the hypothesis but not the conclusion, so the statement is **false**.
why: A single counterexample settles it.
- 3 The converse: if $x = 1$ then $x^2 = x$. Substituting gives $1 = 1$. **True.**
why: Verifying one value is enough because the hypothesis pins x down completely.
- 4 So: false statement, true converse.
why: Match to the option list.

Where it goes wrong

- Dividing $x^2 = x$ by x , getting only $x = 1$, and wrongly declaring the statement true.
- Assuming a false statement must have a false converse - the two are independent.

Takeaway A statement and its converse have no logical connection; test each on its own, and never divide an equation by a variable.

Question 11 PRF-19 · Mathematical proof · spec Prf1

A student proves that $\sqrt{4}$ is irrational by copying the standard proof for $\sqrt{2}$: 'Suppose $\sqrt{4} = p/q$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even, $p = 2k$; then $4k^2 = 4q^2$, so $k^2 = q^2$...' At which point does the argument break down compared with the $\sqrt{2}$ proof?

Answer: C

For $\sqrt{2}$ the step $p^2 = 2q^2$ leads to $2k^2 = q^2$, making q even. Here $k^2 = q^2$ carries no parity information, so the contradiction never appears - as expected since $\sqrt{4} = 2$.

Full method - how and why

Diagnosing why a proof technique fails when transplanted. Follow the argument line by line and find the step that no longer bites.

- 1 Suppose $\sqrt{4} = \frac{p}{q}$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even - this much is **fine**.
why: $4q^2$ is certainly even, and an odd number squared is odd, so p must be even.
- 2 Substituting $p = 2k$: $4k^2 = 4q^2$, i.e. $k^2 = q^2$.
why: Here the 4 cancels entirely, because the constant was 4 and not 2.
- 3 Compare with $\sqrt{2}$: there $p^2 = 2q^2$ gives $4k^2 = 2q^2$, i.e. $q^2 = 2k^2$ - a spare factor of 2 forcing q even.
why: That leftover factor of 2 is the whole engine of the classical proof.
- 4 With $\sqrt{4}$ no factor is left over, so $k^2 = q^2$ tells you nothing about the parity of q , and no contradiction ever appears.
why: Correctly so: $\sqrt{4} = 2$ is rational, so no proof of irrationality could possibly succeed.

Where it goes wrong

- Claiming $p^2 = 4q^2$ fails to make p even - it does not fail; that step is valid.
- Claiming the 'lowest terms' assumption is illegitimate - every rational can be written that way.
- Accepting the argument and concluding $\sqrt{4}$ is irrational, when $\sqrt{4} = 2$.

Takeaway The $\sqrt{2}$ proof works because 2 is prime; the leftover prime factor is what forces the second parity. Check where an adapted proof loses its engine.

Question 12 X2-16 · Differentiation · spec MM6.3

The function f has $f'(x) = (x - 1)(x - 3)^2$. Which of the following statements is/are true?

- I f has a local minimum at $x = 1$.
- II f has a local maximum at $x = 3$.
- III f is increasing for all $x > 1$.

Answer: F

f' changes from negative to positive at $x = 1$ (minimum). At $x = 3$ the squared factor keeps $f' \geq 0$ on both sides, so no turning point: II is false, and $f' \geq 0$ for $x > 1$ (zero only at 3) makes III true.

Full method - how and why

Classify stationary points by whether f' **changes sign** - a squared factor does not change sign.

- 1 $f'(x) = (x - 1)(x - 3)^2$, so $f' = 0$ at $x = 1$ and $x = 3$.
why: Stationary points are the roots of f' .
- 2 Sign of f' : $(x - 3)^2 \geq 0$ always, so the sign is that of $(x - 1)$. Negative for $x < 1$, positive for $x > 1$ (zero at $x = 3$).
why: An even-power factor contributes no sign change.
- 3 I: f' goes negative $\rightarrow$ positive at $x = 1$, so a local **minimum. True.**
why: Falling then rising.
- 4 II: at $x = 3$, f' is positive on both sides, so no turning point - a stationary point of inflexion. **False.**
why: The squared factor is precisely why there is no maximum here.
- 5 III: for $x > 1$, $f' \geq 0$ with equality only at the isolated point $x = 3$, so f is increasing. **True.**
why: An isolated zero of the derivative does not interrupt increase.
- 6 So I and III.
why: Match to the list.

Where it goes wrong

- Assuming every root of f' is a turning point and alternating min/max - the repeated root breaks the alternation.
- Rejecting III because $f'(3) = 0$; increase needs $f' \geq 0$ with no interval of equality.

Takeaway Odd-power factors of f' change sign (turning point); even-power factors do not (stationary inflexion).

Question 13 PRF-12 · Mathematical proof · spec Prf2

For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?

Answer: A

If n were a multiple of 12 it would be a multiple of both 4 and 6, contradicting (i). So by (ii) n is a multiple of 4, and by (i) not a multiple of 6.

Full method - how and why

Deduction from two premises. Test the strongest-looking option against premise (i) first.

- 1 Suppose n were a multiple of 12. Then $12 = 4 \times 3$ and $12 = 6 \times 2$, so n is a multiple of both 4 and 6.

why: Exploring the tempting case first often resolves the whole question.

- 2 But (i) says a multiple of 4 is not a multiple of 6 - contradiction. So n is **not** a multiple of 12.

why: A supposition leading to a contradiction is ruled out.

- 3 Premise (ii) offers two alternatives and the first is now impossible, so n must be a multiple of 4.

why: Disjunctive syllogism: 'A or B' plus 'not A' gives B.

- 4 Then (i) applies: n is a multiple of 4 but not of 6.

why: That is the full deduction.

Where it goes wrong

- Concluding n is a multiple of 12 by picking the first branch of (ii) without testing it.
- Declaring the premises inconsistent - they are perfectly consistent (e.g. $n = 4$ or $n = 8$).
- Concluding nothing, missing that one branch of the 'or' is eliminated.

Takeaway With 'A or B', eliminate one branch and the other must hold; always test the branch that clashes with the other premise.

Question 14 LGC-20 · The logic of arguments · spec Arg1

Which of the following statements is/are logically equivalent to 'if it is raining, then the ground is wet'?

- I It is not raining, or the ground is wet.
- II If the ground is not wet, then it is not raining.
- III If the ground is wet, then it is raining.

Answer: E

' $A \Rightarrow B$ ' is equivalent to 'not A, or B' (I) and to its contrapositive (II). III is the converse, which is not equivalent.

Full method - how and why

Two standard equivalents of $A \Rightarrow B$: the contrapositive, and the disjunctive form 'not A, or B'.

- 1 I: 'not A or B'. An implication only fails when A is true and B false - exactly when ' $\neg A$ or B' fails. **Equivalent.**

why: Check the one bad case: raining and ground dry makes both statements false; every other case makes both true.

- 2 II: 'if the ground is not wet, it is not raining' - the contrapositive. **Equivalent.**

why: Swap and negate.

- 3 III: 'if the ground is wet, it is raining' - the converse. **Not equivalent**; the ground could be wet from a burst pipe.

why: A concrete alternative cause is the cleanest refutation.

- 4 So I and II.

why: Match to the list.

Where it goes wrong

- Rejecting I because it looks nothing like an implication - test it against the single case that breaks an implication.
- Accepting III because wet ground is evidence of rain; evidence is not implication.

Takeaway $A \Rightarrow B$ is equivalent both to 'not A, or B' and to $\neg B \Rightarrow \neg A$. The converse is never among the equivalents.

Question 15 ERR-17 · Identifying errors in proofs · spec Err1

Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

Answer: E

x^4 has $f'(0) = f''(0) = 0$ but a minimum at 0, not an inflexion. (x^3 does have an inflexion; x^2 has $f''(0) \neq 0$.)

Full method - how and why

To disprove a claim about a test for inflexion, produce a function satisfying the conditions but failing the conclusion.

1 Need $f'(a) = f''(a) = 0$ with **no** inflexion at a .

why: Both hypotheses must hold, and the conclusion must fail.

2 Take $f(x) = x^4$ at $a = 0$: $f'(x) = 4x^3$ so $f'(0) = 0$ ✓; $f''(x) = 12x^2$ so $f''(0) = 0$ ✓.

why: Both derivatives vanish - the hypotheses are satisfied.

3 But x^4 has a **minimum** at 0 (it is positive on both sides), not a point of inflexion.

why: An inflexion requires the concavity to change sign; $f'' = 12x^2 \geq 0$ on both sides, so it does not.

4 Check the others: x^3 does have an inflexion at 0 (so it confirms, not refutes); x^2 has $f''(0) = 2 \neq 0$; $x^3 - 3x$ at $a = 1$ has $f'(1) = 0$ but $f''(1) = 6 \neq 0$; $f(x) = x$ has $f'(0) = 1 \neq 0$.

why: Each fails one of the two hypotheses, or satisfies the conclusion.

Where it goes wrong

- Choosing x^3 , the standard **example** of the claim rather than a counterexample.
- Choosing a function where one of the two derivative conditions fails at the stated point.

Takeaway $f''(a) = 0$ does not guarantee an inflexion; the second derivative must **change sign**. x^4 at 0 is the standard counterexample.

Question 16 QUA-17 · Quadratic functions · spec MM1.3

The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I If $ac < 0$, the equation has two distinct real roots.
- II If $b = 0$, the equation has no real roots.
- III If the equation has a repeated root, then $b^2 = 4ac$.

Answer: F

I: $b^2 - 4ac > 0$ whenever $ac < 0$, so true. II: $x^2 - 4 = 0$ has $b = 0$ and real roots, so false. III: a repeated root means the discriminant is zero, so true.

Full method - how and why

Three general claims about $ax^2 + bx + c = 0$. Each is a discriminant statement in disguise; the false one needs a counterexample.

- 1 I: if $ac < 0$ then $-4ac > 0$, so $b^2 - 4ac \geq -4ac > 0$: two distinct real roots. **True.**
why: $b^2 \geq 0$ always, so a positive $-4ac$ forces the whole discriminant positive. Geometrically, a and c of opposite signs means the curve crosses the axis.
- 2 II: test $b = 0$ with $x^2 - 4 = 0$, which has roots ± 2 . **False.**
why: One counterexample settles a 'for all' claim. With $b = 0$ the discriminant is $-4ac$, whose sign depends on a and c .
- 3 III: a repeated root means the discriminant is zero, i.e. $b^2 - 4ac = 0$, i.e. $b^2 = 4ac$. **True.**
why: This is the definition of the repeated-root case, just rearranged.
- 4 I and III only.
why: Match the verdicts to the options.

Where it goes wrong

- Believing II because $b = 0$ 'removes' the linear term and the graph 'sits on the axis' - it does not; $x^2 - 4$ is a counterexample.
- Mis-stating III as $b^2 = -4ac$.

Takeaway Statements about roots translate into statements about $b^2 - 4ac$; to refute one, look for the simplest numbers that fit the hypothesis.

Question 17 TRI-12 · Triangles, radians and sectors · spec MM4.1

A triangle has sides a, b, c with $a^2 + b^2 < c^2$. Which of the following statements is/are true?

- I The angle opposite c is obtuse.
- II c is the longest side.
- III The triangle cannot be isosceles.

Answer: E

Cosine rule: $\cos C = \frac{a^2 + b^2 - c^2}{2ab} < 0$, so C is obtuse (I). The largest angle is opposite the longest side, so c is longest (II). III is false: $a = b = 1, c = 1.5$ works.

Full method - how and why

A condition on the sides. The cosine rule converts it into a statement about an angle; then test the third claim with a counterexample.

- 1 I: $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$, and $a^2 + b^2 < c^2$ makes the numerator negative, so $\cos C < 0$ and C is obtuse. **True.**
why: The denominator $2ab$ is positive, so the sign of $\cos C$ is the sign of $a^2 + b^2 - c^2$.
- 2 II: a triangle can have at most one obtuse angle, and the largest angle is opposite the largest side. Since C is obtuse it is the largest angle, so c is the longest side. **True.**
why: Two obtuse angles would already exceed 180° .
- 3 III: try $a = b = 1$ and $c = 1.5$: then $a^2 + b^2 = 2 < 2.25 = c^2$ ✓ and the triangle is isosceles. **False.**
why: A single counterexample kills the claim. Check the triangle inequality too: $1 + 1 > 1.5$ ✓.
- 4 I and II only.
why: Match to the option list.

Where it goes wrong

- Assuming an obtuse triangle cannot be isosceles.
- Forgetting to check the triangle inequality when constructing a counterexample.

Takeaway $a^2 + b^2 < c^2 \Leftrightarrow$ the angle opposite c is obtuse; obtuse triangles can still be isosceles.

Question 18 X2-09 · Graphs of functions · spec MM8.7

The graphs of $y = f(x)$ and $y = g(x)$ meet at exactly three points. Which of the following statements must be true?

- I The equation $f(x) = g(x)$ has exactly three real solutions.
- II The graph of $y = f(x) - g(x)$ crosses the x -axis exactly three times.
- III $f(x) - g(x)$ is a polynomial of degree at least 3.

Answer: B

I is the definition of an intersection. II fails because the difference could touch the axis rather than cross it at one of the points. III fails because f and g need not be polynomials at all (e.g. $\sin x$ and a line).

Full method - how and why

Intersections and roots. Distinguish a root from a **crossing**, and resist assuming the functions are polynomials.

- 1 I: the graphs meet where $f(x) = g(x)$, so three intersection points means exactly three solutions. **True** - it is the definition.

why: Each intersection point has a distinct x -value, since a graph has one y per x .

- 2 II: at a root of $f - g$ the graph could **touch** the x -axis and turn back rather than cross - a repeated root.

why: Meeting and crossing are different: $y = x^2$ meets the axis but does not cross it.

- 3 So the number of crossings could be fewer than three. **False.**

why: For instance $f - g = x^2(x - 1)$ meets the axis at 0 and 1 but crosses only at 1.

- 4 III: nothing says f and g are polynomials. Take $f(x) = \sin x$ and a suitable straight line g , meeting three times. **False.**

why: Read the hypotheses: only 'the graphs of $y = f(x)$ and $y = g(x)$ ' is given.

- 5 So I only.

why: Match to the list.

Where it goes wrong

- Accepting II by treating 'crosses' as a synonym for 'meets'.
- Accepting III by assuming a degree just because three intersections are mentioned.

Takeaway Intersections = solutions of $f = g$; but crossing the axis needs a sign change, and never assume a function type that was not stated.

Question 19 LGC-05 · The logic of arguments · spec Arg3

Which of the following statements is/are true?

- I For every real number x , there exists a real number y such that $y > x$.
- II There exists a real number y such that for every real number x , $y > x$.
- III For every real number x , there exists a real number y such that $xy = 1$.

Answer: B

I is true (take $y = x + 1$). II claims a single largest real number exists: false. III fails for $x = 0$.

Full method - how and why

Quantifier **order** matters. ' $\forall x \exists y$ ' and ' $\exists y \forall x$ ' say completely different things.

- 1 I: for each x , can I find a bigger y ? Take $y = x + 1$. **True.**
why: y is allowed to depend on x , because x is chosen first.
- 2 II: is there a single y bigger than **every** real x ? That would be a largest real number. Taking $x = y$ contradicts $y > x$. **False.**
why: Here y is fixed **before** x , so it must work for all x at once - including $x = y$.
- 3 III: for each x is there y with $xy = 1$? For $x = 0$, $0 \cdot y = 0 \neq 1$. **False.**
why: 'For every x ' includes the awkward cases; zero has no reciprocal.
- 4 So I only.
why: Match to the list.

Where it goes wrong

- Reading I and II as the same statement - the difference is the order of the quantifiers, nothing else.
- Forgetting $x = 0$ in III and accepting $y = \frac{1}{x}$ uncritically.

Takeaway In $\forall x \exists y$ the y may depend on x ; in $\exists y \forall x$ one y must serve every x . And 'for every' always includes the edge cases.

Question 20 PRF-18 · Mathematical proof · spec Prf2

Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

- I S contains a positive multiple of 5.
- II S contains no positive multiple of 3.
- III S contains a negative number.

Answer: E

$10 \in S$ is a positive multiple of 5 (I). Multiples of 3 in S are negative, so none is positive (II). III fails: $S = \{10\}$ satisfies everything.

Full method - how and why

'Must be true' with a concrete object given. Some claims follow from the given element; others need a counterexample set.

- 1 I: $10 \in S$ and $10 = 2 \times 5$ is a positive multiple of 5. **Must be true.**
why: The element is handed to you; nothing more is needed.
- 2 II: the second condition says every multiple of 3 in S is negative, so none can be positive. **Must be true.**
why: Note this is vacuously fine even if S contains no multiples of 3 at all.
- 3 III: must S contain a negative number? Try $S = \{10\}$: every element is a multiple of 3 or 5 ✓, there are no multiples of 3 to be negative ✓, and $10 \in S$ ✓. No negative number. **Not necessarily true.**
why: A single valid S with no negative element refutes 'must be true'.
- 4 So I and II.
why: Match to the list.

Where it goes wrong

- Reading 'every multiple of 3 in S is negative' as promising that S **has** a multiple of 3 - a universal statement can be true with nothing to apply it to.
- Confusing 'could be true' with 'must be true' for III.

Takeaway A conditional statement about members of a set is vacuously true when no member qualifies; to refute 'must be true', build a legal example that fails it.